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original article

Cooled radiofrequency ablation for patients with chronic postsurgical pain after total knee arthroplasty: Pilot clinical trial

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ABSTRACT

Introduction: Total knee arthroplasty (TKA) generally provides substantial and durable relief in end-stage knee osteoarthritis, yet a proportion of patients develops chronic postsurgical pain (CPSP). Conventional analgesics often fail to adequately address this condition. Cooled radiofrequency ablation (CRFA) has shown effectiveness in knee osteoarthritis, but its utility for patients with CPSP after TKA has not been investigated. This study aimed to report a phase I/IIa clinical trial regarding CRFA for patients with CPSP after TKA.

Materials and methods: This prospective pilot study (jRCTs052230123) enrolled consecutive patients with CPSP between November 2023 and August 2025. Inclusion criteria included persistent pain ≥6 months after TKA, inadequate response to standard analgesics (NRS ≥5), and ≥50% pain reduction after diagnostic genicular nerve block. Following informed consent, participants underwent CRFA using the Coolief® system. The primary endpoint was the change in Numeric Rating Scale (NRS) pain score from baseline to 6 months. 2011 Knee Society Score (KSS) was also used for the efficacy evaluation. As for the safety evaluation, the occurrence and outcomes of any severe adverse events between the day of treatment and 6 months post-treatment were recorded.

Results: Of 15 screened patients, 5 met all criteria and completed the study. CRFA resulted in a consistent reduction in pain intensity; however, the change in NRS scores did not reach statistical significance at 6 months. Responder analyses demonstrated that a substantial proportion of patients achieved clinically meaningful improvements at 6 months. Although the 2011 KSS showed a trend toward improvement, the change was not statistically significant. No adverse events or complications occurred.

Conclusions: CRFA provided clinically meaningful pain relief without adverse events in patients with CPSP after TKA. These findings suggest that CRFA is a safe and potentially effective therapeutic option. Larger studies are needed to verify its clinical value.

1. Introduction

Total knee arthroplasty (TKA) is an established and effective treatment for end-staged knee osteoarthritis (KOA) that substantially improves ambulation and quality of life. Long-term follow-up studies have reported favorable outcomes lasting over two decades, confirming its reliability as a durable surgical procedure.[1] Despite these successful outcomes, a consistent proportion of patients experiences chronic postsurgical pain (CPSP) following TKA. Western studies have reported that CPSP develops in approximately 20% of cases after TKA.[2] Peripheral nerve injury during surgery and central sensitization are thought to be major contributing mechanisms.[3]

Conventional pharmacologic treatments—including Non-Steroidal Anti-Inflammatory Drugs (NSAIDs), corticosteroids, opioids, and pregabalin—often fail to provide adequate relief. Although newer agents have extended therapeutic options, none have demonstrated consistent efficacy in patients with CPSP after TKA.[4] Radiofrequency ablation (RFA) of the genicular nerves has been used to manage pain in patients with KOA.[5] This procedure disrupts nociceptive transmission by inducing axonal injury in sensory nerve branches of the joint capsule. However, traditional RFA is limited by tissue carbonization at high

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Abbreviations

(TKA) total knee arthroplasty
(KOA) knee osteoarthritis
(CPSP) chronic postsurgical pain
(RFA) radiofrequency ablation
(CRFA) cooled radiofrequency ablation
(KOOS) Knee Injury and Osteoarthritis Outcome Score
(WOMAC) Western Ontario and McMaster Universities Osteoarthritis Index
(NRS) numeric rating scale
(KSS) Knee Society Score
(ROM) range of motion

temperatures, which restricts lesion size and reduces effectiveness. The cooled radiofrequency ablation (CRFA) system (Coolief®) incorporates an internal cooling mechanism that minimizes carbonization and enables the creation of larger, more uniform thermal lesions, potentially resulting in superior clinical outcomes.[6] While the efficacy and safety of CRFA in patients with KOA have been well documented,[7–9] few detailed reports have evaluated CRFA for patients with CPSP after TKA. A recent study by Khan et al. evaluated CRFA for chronic pain after TKA and demonstrated favorable outcomes in pain and function.[10] In their report, pain relief was assessed using the Visual Analog Scale, and functional improvement was evaluated using the Knee Injury and Osteoarthritis Outcome Score (KOOS) and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), with a 10.2-month follow-up. Furthermore, while CRFA for KOA typically involves ablation of three knee nerve branches (superomedial, superolateral, and inferomedial branches), this study also included ablation of the suprapatellar nerve in addition to the three primary nerves. Additionally, this study was retrospective and did not clearly define the study population as patients with CPSP. Therefore, prospective data focusing on a clearly defined CPSP population remains limited.

This study aimed to exploratory evaluate the efficacy and safety of CRFA for the management of patients with CPSP after TKA. We hypothesize that CRFA treatment would yield superior clinical outcomes without any severe adverse events, particularly regarding pain, in patients with CPSP after TKA.

2. Materials and Methods

2.1. Study design and sample size

This prospective pilot study was registered in the Japan Registry of Clinical Trials (jRCTs052230123). Patients were prospectively enrolled from consecutive individuals who presented to our institution between November 1st, 2023, and August 12th, 2025, and met the predefined eligibility criteria. Trained investigators screened all participants in accordance with the study protocol. After confirming eligibility, written informed consent was obtained prior to any study-related procedures. Patients were then registered sequentially and followed according to the prespecified assessment schedule. It was estimated that at least 5 participants would need to enroll during the planned registration period. Considering the possibility of dropouts during the study period, the target sample size was set at 6 participants. No formal sample size calculation was performed due to its exploratory design. The sample size was determined based on feasibility, expected recruitment during the study period, and the primary objective of assessing safety and preliminary signals of efficacy rather than hypothesis testing. Therefore, the number of participants was not intended to provide sufficient statistical power but to inform the design of future larger-scale studies. This study was designed as a single-arm prospective pilot study without a control group. The absence of a control group was primarily due to ethical and practical considerations. CRFA is an invasive procedure requiring local anesthesia and ultrasound-guided probe placement, making the implementation of a sham-controlled design challenging. In addition, withholding treatment from patients with persistent moderate-to-severe pain after TKA was considered ethically difficult. Therefore, this study was conducted as a single-arm trial to primarily evaluate safety and preliminary efficacy.

2.2. Eligibility criteria, consent to participate, and participation timeline

Prior to inclusion, all patients underwent comprehensive clinical and radiographic evaluation to exclude mechanical causes of pain, including prosthetic loosening, malalignment, instability, or infection. Each participant was confirmed to meet all inclusion criteria and not meet any exclusion criteria before enrollment (Screening). Baseline information, including date of surgery, age, sex, body mass index, medical history, and current medications, was collected. All eligible participants are required to provide written informed consent. The detailed schedule is shown in Table 1.

Inclusion criteria are listed below;

1. Patients with chronic knee pain persisting for ≥6 months after TKA.
2. Patients aged 18 years or older, regardless of sex.
3. Patients who did not achieve satisfactory pain relief (NRS score ≥5) despite conventional analgesic therapies such as NSAIDs, steroids, opioids, or pregabalin.
4. Patients who demonstrated at least a 50% reduction in NRS score following diagnostic genicular nerve block (superomedial, superolateral, and inferomedial branches). This block has been defined as a test block.
5. Patients who voluntarily provided written informed consent to participate in this clinical study.

Exclusion criteria are listed below;

1. Patients with inflammatory conditions in the knee (e.g., septic arthritis, gout, pseudogout).
2. Patients with systemic infections.
3. Patients with neurological disorders such as diabetic peripheral neuropathy, syringomyelia, or spinal fistula.
4. Patients with bleeding disorders or those receiving anticoagulant therapy.
5. Patients who had previously undergone CRFA for KOA.
6. Patients with implanted pacemakers or defibrillators.
7. Patients deemed inappropriate for inclusion by the investigators.

2.3. Interventions

The device name is an electrosurgical unit for ablation — Coolief Cooled Radiofrequency Pain Management System (Avanos Medical Japan, Inc.) (Fig. 1a). The following procedures were performed as part of the trial:

1. Under ultrasound guidance, a Cooled RF introducer (hereinafter referred to as the “introducer”) is inserted into the target biological tissue (superomedial, superolateral, and inferomedial genicular nerves) (Fig. 1b).
2. The stylet is removed from the cannula, and the probe is inserted.
3. The electrical stimulation mode (“Stimulation”) is selected.
4. The desired probe channel is chosen.
5. The “Sensory 50 Hz” mode is selected using the stimulation mode button.
6. The start button is slid to the right to initiate the stimulation mode, which is maintained for 2 min and 30 s per nerve (Fig. 1c and d).
7. The stop button is pressed to terminate electrical stimulation.

Table 1
Participation timeline.

Items	Timing					
	Screening period	Treatment	Follow-up period			At discontinuation
			1 month	3 months	6 months	
Acceptable range (days)	−28	±0	±5	±10	±15	
Consent acquisition	●					
Case registration	●					
Patient background data acquisition	●	●				
Device use		●				
Numeric rating scale	●		●	●	●	●
2011 Knee Society Score	●		●	●	●	●
Confirmation of adverse events		●	●	●	●	●

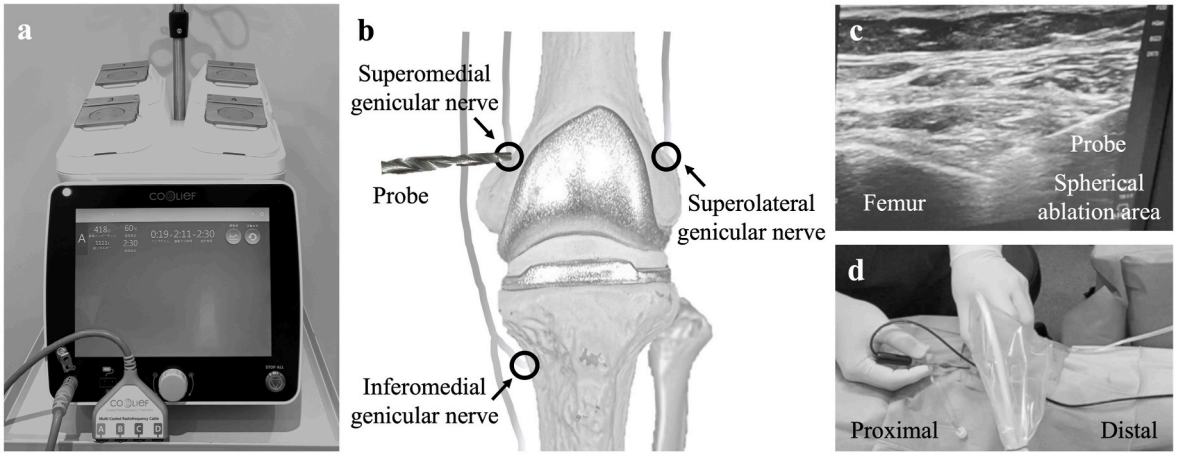


Fig. 1. a. Coolief Cooled Radiofrequency Pain Management System b. Target nerves c and d. Performing ablation of the superomedial genicular nerve on the left knee While monitoring the echo, advance the probe tip toward the ablation site. During the actual ablation procedure, the spherical area can be visible.

All procedures were performed by a single experienced operator (M. T.) using standardized anatomical landmarks under ultrasound guidance to ensure consistency and reproducibility. The anatomical landmarks used in this study were as follows:

Superomedial genicular nerve: Identified at the intersection where the femoral shaft meets the medial epicondyle, located superior and medial to the knee joint.

Superolateral genicular nerve: Located at the intersection of the femoral shaft and the lateral epicondyle, found superior and lateral to the knee joint.

Inferomedial genicular nerve: Determined at the intersection of the tibial shaft and the medial condyle, positioned inferior and medial to the knee joint.

The cost of the CRFA system consists of both capital equipment and disposable components. The manufacturer's suggested retail price in Japan is approximately ¥7.5 million for the generator and pump unit combined, and approximately ¥149,000 per disposable probe. However, it should be noted that the actual acquisition cost may be lower than the manufacturer's suggested retail price due to institutional pricing agreements and bulk purchasing arrangements.

2.4. Endpoints

The primary endpoint is a comparison of the NRS score[11] for pain between prior to treatment (baseline) and 6 months after treatment. NRS scores were assessed during movement (weight-bearing activity), as this was considered most representative of patients' functional pain.

Secondary endpoints consist of efficacy and safety endpoints. For efficacy evaluation, subscales of the 2011 Knee Society Score (KSS) and range of motion (ROM) of the knee were compared between prior to treatment and 6 months after treatment. Additionally, we conducted longitudinal assessments of the NRS score and the 2011 KSS (baseline, and 1-, 3-, and 6-month after treatment). As for the safety evaluation, the occurrence and outcomes of any severe adverse events potentially related to the investigational device were recorded during the day of treatment to 6 months post-treatment. Adverse events to be monitored during the 6-month follow-up period include, but are not limited to: burns; seizures or muscle contractions; bleeding; infections; nerve injury; increased pain; visceral injury; gastrointestinal disorders; respiratory impairment; paralysis; death; and sensory disturbances due to afferent nerve involvement. Information regarding concomitant analgesic use was recorded at baseline and during follow-up visits.

2.5. Statistical analysis

All values are expressed as mean ± standard deviation. Data analyses were performed using a statistical software package (Statview 5.0; Abacus Concepts Inc., Berkeley, CA). The Shapiro-Wilk test (SPSS Statistics 21; IBM Japan, Tokyo, Japan) was performed to analyze normally distributed data. Due to the small sample size, non-parametric statistical methods were used. Continuous variables are also presented as median and interquartile range. For the primary endpoint, the change in NRS scores between baseline and 6 months after treatment was analyzed using the Wilcoxon signed-rank test. For the secondary endpoint, the Wilcoxon signed-rank tests were also applied to compare the 2011 KSS and ROM between baseline and 6 months.

To assess longitudinal changes over time, including baseline and 1-, 3-, and 6-month evaluations, one-way analysis of variance was performed for both the NRS score and 2011 KSS. When significant differences were identified, post hoc pairwise comparisons were performed

using the Wilcoxon signed-rank test with Bonferroni correction.

In addition to statistical testing, descriptive analyses including individual change scores and responder rates (≥2-point, ≥30%, and ≥50% improvement in NRS) were reported to enhance clinical interpretability. Based on previous reports indicating that a reduction of ≥2 points on the NRS represents a clinically important improvement in chronic pain.[12] P values of <0.05 were considered to be statistically significant.

3. Results

3.1. Participants

In this study, 15 candidates were initially considered; however, 6 participants were enrolled as planned because the test block did not demonstrate at least a 50% reduction in NRS scores (inclusion critera No. 4). In addition, after case registration, it was discovered that one patient had a history of diabetic peripheral neuropathy (exclude criteria No. 3). Therefore, a total of 5 participants were ultimately included in the study. Patient characteristics were listed in Table 2. All included patients demonstrated a ≥50% reduction in NRS following diagnostic genicular nerve block, confirming the presence of a peripheral pain component. One patient was initially enrolled but was later excluded after detailed chart review by a third party (the Institutional Review Board) revealed a history of diabetic peripheral neuropathy, which had not been identified during the initial screening process.

Table 2
Participant characteristics.

No	Age	Sex	Body mass index (kg/m^2)	NRS baseline	NRS after block	Date of surgery	Types of implants and surgical approaches	Date of intervention	Duration (months)	Medical history	Current medications
1	78	Female	30.3	7	3	2021/6/19	JOURNEY II BCS (Smith & Nephew, Memphis, TN, USA) and medial parapatellar approach	2024/1/9	30	- Hypertension - Chronic Allergies	Duloxetine hydrochloride 30 mg Domperidone 10 mg Olmesartan medoxomil 5 mg Amlodipine besylate 2.5 mg Cetirizine hydrochloride 10 mg
2	69	Female	23.8	5	2	2021/10/6	Vanguard PS (Zimmer Biomet, Warsaw, IN, USA) and medial parapatellar approach	2024/2/26	28	- Bronchial asthma - Hyperlipidemia - Chronic sinusitis	Diphenidol hydrochloride 25 mg Fexofenadine hydrochloride 60 mg Vonoprazan fumarate 10 mg Fluvastatin sodium 30 mg Carbocisteine 500 mg Clarithromycin 200 mg
3	89	Female	25.2	7	3	2016/11/8	Vanguard PS (Zimmer Biomet, Warsaw, IN, USA) and medial parapatellar approach	2024/3/25	88	- Hypertension - Lumbar Spinal - Canal Stenosis	Pitavastatin calcium 2 mg Azosemide 30 mg Amlodipine besylate 2.5 mg Sacubitril and valsartan 50 mg Verapamil hydrochloride 40 mg Tocopherol acetate 50 mg Mirogabalin besylate 5 mg Acetaminophen 500 mg
4	80	Female	28.9	8	4	2012/5/17	OPTETRAK PS (Exactech, Gainesville, FL, USA) and medial parapatellar approach	2024/11/11	149	- Hypertension - Hyperlipidemia - Insomnia	Indapamide 1 mg Atorvastatin calcium 10 mg Eszopiclone 1 mg
5	77	Female	28.5	9	4	2020/8/4	Persona PS Knee (Zimmer Biomet, Warsaw, IN, USA) and medial parapatellar approach	2025/5/26	57	- Hypertension - Hyperlipidemia - Breast Cancer	Amlodipine besylate 5 mg Ezetimibe 10 mg Eszopiclone 1 mg Diazepam 5 mg *Clostridium butyricum* 10 mg

NRS; numeric rating scale, PS; Posterior-Stabilized.

3.2. Clinical results

Table 3 shows detailed data on the NRS score, the 2011 KSS, and ROM at baseline and 6 months after treatment. At 6 months post-treatment, NRS scores trended toward improvement, but the difference was not statistically significant. Similarly, trends toward improvement were seen in the symptoms and functional activity subscales of the 2011 KSS, but these changes were also not statistically significant (Figs. 2 and 3). ROM showed no significant change between baseline and 6 months. Table 4 shows longitudinal changes in clinical data over time, including baseline and 1-, 3-, and 6-month evaluations. The NRS score showed significant improvement at 1 month after treatment compared with prior to treatment. Functional activities scores showed significant improvement at 6 months after treatment compared with baseline (Table 4).

Although the reduction in NRS scores did not reach statistical significance using the Wilcoxon signed-rank test, a consistent improvement was observed in most patients. Specifically, responder analysis of the NRS score demonstrated that 5/5 patients (100%) achieved a ≥2-point reduction, and 4/5 (80%) achieved ≥30% improvement, while 2/5 (40%) achieved ≥50% improvement at 1 month after treatment, 4/5 patients (80%) achieved a ≥2-point reduction, and 3/5 (60%) achieved ≥30% improvement, while 3/5 (60%) achieved ≥50% improvement at 3 months after treatment, and 4/5 patients (80%) achieved a ≥2-point reduction, and 4/5 (80%) achieved ≥30% improvement, while 2/5 (40%) achieved ≥50% improvement at 6 months after treatment (Table 5 and Fig. 4).

Finally, no substantial changes in analgesic use or concomitant therapies were observed during the follow-up period, and no adverse events or complications were reported during the study period.

4. Discussions

In the present study, CRFA resulted in a consistent reduction in pain intensity, although the change in NRS scores did not reach statistical significance at 6 months when analyzed using non-parametric methods. Importantly, responder analyses demonstrated that a substantial proportion of patients achieved clinically meaningful improvements, with 80% achieving ≥30% reduction and 40% achieving ≥50% reduction at 6 months. These findings suggest that CRFA may provide clinically relevant pain relief even in the absence of statistical significance, particularly in small exploratory cohorts.

CRFA was developed as a significant advancement over conventional RFA for the treatment of KOA. The key technological advantage of CRFA is its internally cooled electrode, which prevents tissue carbonization at the probe–tissue interface.[13] This allows continuous energy delivery, maintains low impedance, and produces larger, more spherical, and more uniform lesions than conventional RFA, where high temperatures often result in charring and limit lesion size.[5] Larger lesion volumes improve the likelihood of capturing anatomically variable genicular nerve branches, thereby enhancing the completeness of neurotomy and the durability of analgesia.[8] Clinical evidence also supports the benefit of CRFA. A randomized, sham-controlled trial demonstrated significantly greater pain reduction with CRFA, with approximately 75% of KOA patients achieving ≥50% improvement at 3 months.[6] Retrospective analyses similarly show substantial and sustained reductions in pain with a favorable safety profile.[7] CRFA offers a mechanistically sound and clinically effective alternative to conventional RFA, with the potential for more reliable pain relief in patients with KOA.

CPSP after TKA represents one of the most challenging frontiers in arthroplasty treatment.[14] Although TKA has long been regarded as a definitive and durable solution for end-stage KOA, a consistent proportion of patients continue to suffer from persistent postoperative pain that compromises both patient satisfaction and functional recovery.[15,16] The present study provides clinical evidence that the CRFA may be an effective and safe interventional therapy for patients with CPSP after

Table 3
Clinical data and comparison between baseline and 6 months after treatment.

Patient number	NRS scores		Symptoms		Patient satisfaction		Patient expectations		Functional activities		Extension angle		Flexion angle	
	Baseline	After treatment	Baseline	After treatment	Baseline	After treatment	Baseline	After treatment	Baseline	After treatment	Baseline	After treatment	Baseline	After treatment
1	7	6	10	8	14	14	12	3	63	65	0	0	100	115
2	5	2	12	15	14	16	8	6	64	77	0	0	100	100
3	7	4	8	13	14	14	7	8	27	55	0	0	100	110
4	8	4	5	13	14	14	8	8	40	55	0	0	110	110
5	9	6	2	8	8	18	8	6	22	59	−5	−5	85	90
Mean ± SD	7.2 ± 1.5	4.4 ± 1.7	7.4 ± 4.0	11.4 ± 3.2	12.8 ± 2.7	15.2 ± 1.8	8.6 ± 1.9	6.2 ± 2.0	43.2 ± 19.7	62.2 ± 9.2	−1.0 ± 2.2	−1.0 ± 2.2	99.0 ± 8.9	105.0 ± 10.0
Median [IQR]	7 1	4 2	8 5	13 5	14 0	14 2	8 0	6 2	40 36	59 10	0 0	0 0	100 0	110 10
Change score	−2.8		4.0		2.4		−2.4		−19.0		0		6.0	
P value	0.06		0.13		0.50		0.25		0.06		-		0.25	

※:statistically significant.
NRS; numeric rating scale, SD; standard deviation, IQR; Interquartile Range.

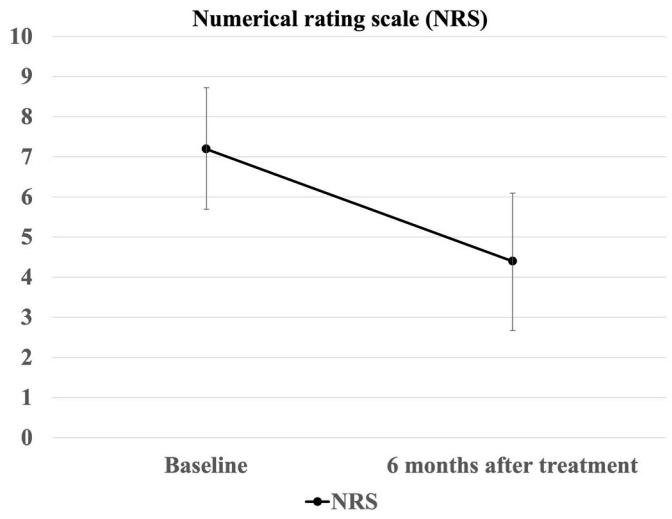


Fig. 2. Change in numeric rating scale (NRS) scores between baseline and 6 months after treatment
No statistically significant improvement in NRS score was observed 6 months after treatment.



Fig. 3. Change in the 2011 Knee Society Score (KSS) between baseline and 6 months after treatment
There was no significant difference in subscales of the 2011 KSS between baseline and 6 months after treatment.

TKA—an area in which no standardized treatment has yet been established.

The responder analyses demonstrated that a substantial proportion of patients achieved clinically meaningful improvements in NRS scores over 1- to 6-month after CRFA, which underscores the critical role of persistent peripheral nociceptive input in maintaining CPSP. Traditionally, CPSP after TKA has been attributed mainly to central sensitization, but our data suggest that peripheral generators—particularly residual nociceptive signaling from capsular or periosteal nerve branches—remain critical contributors in some patients.[17–19] In this study, not all CPSP patients achieved a 50% or greater reduction in NRS scores with the test block. The pathophysiology of CPSP following TKA is multifactorial. Current evidence suggests that persistent pain may arise from postoperative inflammatory processes, injury to peripheral nerve structures, and alterations in neurobiological pain modulation pathways.[20] Some reports suggest that neuropathic pain originating from peripheral neuropathy accounts for approximately 20% of cases of CPSP.[3] These complexities underscore that CPSP does not arise from a single etiology but rather from the interaction of multiple peripheral and central mechanisms. CRFA, by producing broad and controlled thermal neurotomy of the genicular branches, appears to effectively disrupt this peripheral neuropathway. Accordingly, a treatment that can selectively modulate peripheral nociceptive pathways may be effective for patients, especially those in whom peripheral drive remains a dominant source of pain.

Longitudinal changes in NRS score over time showed significant improvement at 1 month after treatment compared with prior to treatment. However, no significant difference was observed at 3 or 6 months after treatment. CRFA produces neurotomy by maintaining tissue temperatures near 77-80°C, creating a large, spherical lesion while preserving the collagenous architecture of the nerve sheath.[21] This temperature reliably induces Wallerian degeneration of targeted sensory axons, leading to rapid interruption of nociceptive signaling and the marked early reduction in knee pain frequently observed in patients with KOA.[22] However, because CRFA does not entirely destroy the epineurial and perineurial scaffold, progressive axonal regeneration is biologically expected. Preclinical and clinical studies show that peripheral sensory nerves begin regenerating within weeks and may regain functional connectivity over several months following thermal ablation, offering a mechanistic explanation for the gradual return of pain or its partial worsening after an initial period of robust relief.[23,24] When these mechanisms are considered collectively, CRFA produces clinically significant pain relief. However, worsening NRS scores observed during the 1- to 3-month period after treatment may reflect the anticipated biological process of nerve repair.

When comparing our findings with those of Khan et al., who reported sustained functional improvement using CRFA with a mean follow-up of 10.2 months, several important differences should be considered.[10] First, differences in targeted nerve branches may have influenced

Table 4
Longitudinal changes of clinical data over time, including baseline and 1-, 3-, and 6-month evaluations.

Items	Timing 0	Timing 1	Timing 2	Timing 3	P value					
	Baseline	1 month after treatment	3 months after treatment	6 months after treatment	Timing 0 vs 1	Timing 0 vs 2	Timing 0 vs 3	Timing 1 vs 2	Timing 1 vs 3	Timing 2 vs 3
NRS scores	7.2 ± 1.5	4.2 ± 1.8	4.4 ± 2.1	4.4 ± 1.7	**0.04**※	0.12	0.12	>0.99	>0.99	>0.99
Symptoms	7.4 ± 4.0	12.8 ± 3.7	12.0 ± 3.8	11.4 ± 3.2	0.16	0.67	>0.99	>0.99	>0.99	>0.99
Patient satisfaction	12.8 ± 2.7	18.4 ± 3.6	17.2 ± 4.8	15.2 ± 1.8	0.08	0.85	>0.99	>0.99	0.85	>0.99
Patient expectations	8.6 ± 1.9	5.8 ± 2.2	6.4 ± 2.6	6.2 ± 2.0	>0.99	>0.99	>0.99	>0.99	>0.99	>0.99
Functional activities	43.2 ± 19.7	46.8 ± 16.0	51.4 ± 9.2	62.2 ± 9.2	>0.99	>0.99	**0.04**※	>0.99	0.09	0.16

※:statistically significant.
The data are given as mean ± standard deviation.
NRS; numeric rating scale.

Table 5
Responder analysis of the NRS score Gray-colored cells indicate either a ≥2-point reduction or a ≥50% improvement.

Patient number	Baseline	1 month after treatment	Change score	% improvement	3 months after treatment	Change score	% improvement	6 months after treatment	Change score	% improvement
1	7	4	3	43	6	1	14	6	1	14
2	5	2	3	60	2	3	60	2	3	60
3	7	3	4	57	3	4	57	4	3	43
4	8	6	2	25	4	4	50	4	4	50
5	9	6	3	33	7	2	22	6	3	33

NRS; numeric rating scale.

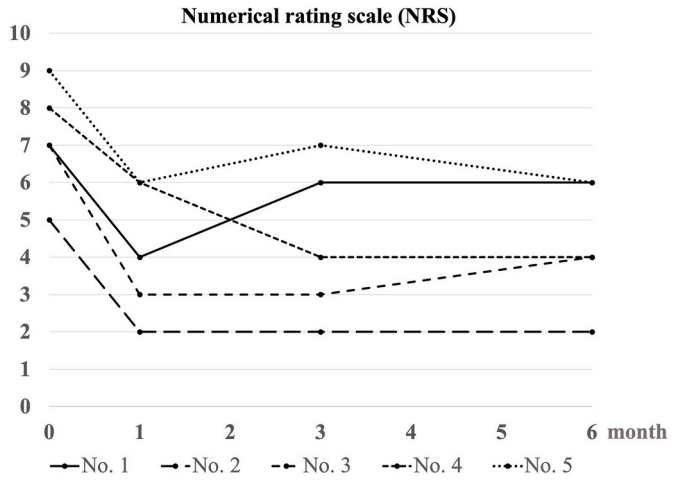


Fig. 4. Individual patient trajectories of numeric rating scale (NRS) scores
Responder analyses demonstrated that a substantial proportion of patients achieved clinically meaningful improvements, with 80% achieving ≥30% reduction and 40% achieving ≥50% reduction at 6 months.

outcomes. Khan et al. included four genicular branches, including the suprapatellar nerve, whereas our study targeted three branches. A broader ablation field may lead to more comprehensive denervation and potentially greater functional benefit. Second, differences in patient characteristics may have contributed to the observed discrepancies. Variations in age, time since TKA, and underlying pain phenotype—particularly the relative contribution of nociceptive versus neuropathic mechanisms—may significantly influence treatment response. Our study specifically focused on patients with CPSP, a population that may have more complex pain mechanisms than those included in previous reports. Third, differences in outcome measures should be considered. While our study used the 2011 KSS, Khan et al. employed KOOS and WOMAC, which may be more sensitive to changes in patient-reported outcomes and functional status.[25,26] The limited changes observed in KSS subscales in our study may therefore reflect differences in measurement sensitivity rather than the true absence of functional improvement.

In addition to therapeutic effects, CRFA in patients with KOA has demonstrated a favorable safety profile in both preclinical and clinical contexts. Multiple studies have reported that serious adverse events are rare following CRFA, with most complications limited to transient discomfort at the treatment site, minor sensory changes, or brief local inflammation that resolve without long-term sequelae.[27,28] In this study as well, no adverse events or complications were observed in any case, confirming that the procedure is safe even after TKA.

Despite the promise shown, there are some limitations in this study. First, the sample size was small, limiting subgroup analyses and reducing statistical power for functional outcomes. Second, the absence of a control group also makes it difficult to exclude placebo effects or natural recovery. However, given that these patients had persistent pain for more than 6 months—and had already failed conventional medical therapies—spontaneous improvement seems unlikely. In addition, the requirement for a ≥50% reduction in pain following diagnostic genicular nerve block supports the presence of a peripheral pain component and strengthens the rationale for CRFA. A sham-controlled design would be methodologically desirable but poses ethical and practical challenges given the invasive nature of the procedure. Third, functional outcomes were assessed using the 2011 Knee Society Score, which may be less sensitive to change than patient-reported measures such as KOOS or WOMAC. Fourth, the cost of the CRFA system may represent a potential barrier to widespread adoption, and formal cost-effectiveness analyses are warranted in future studies. Finally, the follow-up period was limited, and longer-term outcomes remain unclear. Future studies should incorporate randomized controlled trials, larger cohorts, and longer follow-up to assess the durability of CRFA in CPSP patients after TKA.

5. Conclusions

Clinically meaningful improvements in the NRS score was observed 6 months after CRFA, without any adverse events or complications. The present study provides early evidence that CRFA can meaningfully reduce CPSP after TKA with a favorable safety profile. By effectively modulating persistent peripheral pain generators, CRFA may fill an important therapeutic gap in arthroplasty treatment and offer a promising new approach to improving quality of life for patients who remain limited by CPSP. However, these findings should be interpreted cautiously given the small sample size and absence of a control group. Further large-scale, controlled studies are required to confirm the efficacy and cost-effectiveness of CRFA in this population.

Ethics approval statement

We confirm that ethical approval was obtained for this prospective clinical trial. The study was reviewed and approved by the Kobe University Clinical Research Ethical Committee on October 5, 2023. The study was registered in the Japan Registry of Clinical Trials (jRCTs052230123), and written informed consent was obtained from all participants prior to enrollment.

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Conflict of interest statement

The author(s) have no conflicts of interest relevant to this article.

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